Alfred Nonis

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EDUCATION

University of Texas at Austin - B.S., Mechanical Engineering August 2023 - December 2027

Specializing in Design & Manufacturing and Energy Systems

Relevant Coursework: Combustion Engine Processes, Materials Processing, Heat Transfer, Machine Elements, Mechatronics, Materials, Mechanics of Solids, Fluid Dynamics, Matlab, Statistics

EXPERIENCE

Longhorn Baja Racing SAE

Chassis Co-Lead, Founding Member June 2025 - PRESENT

- Co-designed a **parametric roll cage** in SolidWorks through **12 iterations**, integrating suspension, powertrain, and ergonomics while maintaining SAE rule compliance.
- **Optimized load paths** and tubing selection, **reducing roll cage weight by 31%** through member removal, tube downsizing, and structural analysis.
- Performed **Ansys FEA** for 10,000 lbf frontal/rear loads, 1,400 lbf suspension loads, and engine mount loading, evaluating **factors of safety** (~1.3 for the front and 1.5 for the rear) and displacement to guide chassis redesign.
- Applied **Design for Manufacturing (DFM)** principles by redesigning tube joints for single-plane notching and easier jigging. Fabricated tubes for all subsystems, and TIG tack welded primary chassis tubes.

Texas Spacecraft Laboratory

REACT Research, Internal Hardware Lead April 2025 - PRESENT

- Leading a small hardware team, coordinating mechanical design requirements with physics and electronics teams for CubeSat magnetic interference testing.
- Designed and **3D-printed** sensor mounts, snap-fit fasteners, and damping legs in SolidWorks, addressing magnetic interference and fit tolerances through iterative prototyping.
- Developed a **viscous damping system** through **4 design iterations**; currently designing a non-ferromagnetic cooling solution for electronic components during testing.

Agentic Innovations

AI Engineering Intern May 2026 - August 2026

- Co-developed a **multi-agent AI patent analysis application** using Codex, CrewAI, RAG, semantic search, and the **USPTO API** to retrieve and verify patents, generating reports identifying novel features and potential overlaps.
- Co-developed CASE, an **agentic AI BOM sourcing platform** that processes engineering data, evaluates supplier manufacturing capabilities, compares material availability, and generates evidence-backed sourcing reports.
- Used Codex to create CASE's **system architecture** and developed **validation test cases** to verify supplier capabilities and refine sourcing criteria; **selected from ~10 teams** to pitch CASE to engineers from Toyota, NVIDIA, Apple, and the NSA.

Encotech Engineering Consultants, Inc.

MEP Intern June 2022 - August 2022

- Developed electrical outlet layouts and interior floor plans in **Autodesk Revit**, comparing designs against existing engineering drawings.
- Reviewed plumbing BOMs for accuracy and shadowed engineers during HVAC system design.

PROJECTS

Internal Combustion Engine Modification + Engine Kit Build

- Independently designed and 3D-printed a **Helmholtz-tuned** intake velocity stack for a Predator 79cc engine, achieving a **7.2% average torque** increase during testing. Also assembled a commercial 2-stroke engine kit.

SKILLS

Engineering Software: SolidWorks, Ansys, Matlab, Autodesk Inventor, Revit, ideaMaker, Bambu Studio

Hands-on: Worked on brake calipers, pads, rotors, spark plugs, oil/air filters, tire rotation/replacement, EVAP, woodworking, laser cutting, manual hole saw, **TIG welding,** drop saw, band saw

Office Software: MS Word, MS Excel, MS PowerPoint

INVOLVEMENT

Flagging & Communications Marshal for SCCA and the Circuit of the Americas August 2025 - PRESENT